

eXtended Reality Vest: A New Approach to Demonstration-Based Learning

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ABSTRACT

This study examines the design and usability of a new extended reality (XR) training device, XR Vest, which enhances demonstration-based training (DBT) by combining two viewpoints (first- and third-person views). Traditional DBT methods can lack engagement and pose hazards, leading to poor retention. XR technology is used to improve engagement, safety, and immersion. The XR Vest, worn by a trainer, allows for a first-person view of immersive XR environments on an integrated screen while also displaying the trainer's movements in third-person. 28 participants completed training sessions using the XR Vest and a computer monitor. NASA Task Load Index questionnaires and System Usability Scale questionnaires were used to evaluate the effectiveness of the XR Vest. Results showed the XR Vest decreased cognitive load and improved usability. Further research should investigate the benefits of this design in other areas using XR technology for DBT.

Index Terms: Hardware—Emerging Technologies—Analysis and design of emerging devices and systems—Emerging tools and methodologies

1 INTRODUCTION

Jobs come in various levels of complexity. As the complexity of the job increases, so must the training. A popular and effective solution to this comes in the form of demonstration-based training (DBT) [2, 7], which utilizes the basic learning technique of observation and imitation [4]. As jobs become more complex with specialized equipment, the mobility of DBT decreases as it relies on physical equipment. However, the possibility of virtual reality (VR) grants the unique ability for safe and mobile DBT even in simulations of hazardous conditions [6]. With this layer of safety, it is possible to simulate conditions that could be dangerous if performed live and can give immediate safe feedback if the trainee performs a potentially dangerous action [6]. The limitations of using VR head-mounted displays (HMD) for DBT arise when there is a large group of trainees and a minimal number of HMDs [5]. One solution is to instruct the trainee one at a time. However, this could become very costly due to the required time to train the group. Another solution that would still give the benefits of DBT and the flexibility offered by VR is to have the trainer run the simulation in VR and wear a screen for the trainees to observe. With this novel approach, the trainees observe the trainer's movements and watch a screen on the trainer's back to give context to the movements and actions.

This study aims to evaluate the effectiveness of (1) a portable and wearable XR-based platform for DBT that combines live (third-person) and video-based (first-person) perspectives, compared to (2) the XR computer monitor platform. We developed a novel XR Vest that allows trainers to provide trainees with both a live third-person perspective of their motor movements and a first-person view from an integrated screen on the vest. We hypothesize that the design of



Figure 1: Demonstration and design of XR Vest for first- and third-person DBT.

the XR Vest will improve cognitive load and system usability during a DBT task compared to traditional pre-recorded videos.

2 METHOD

2.1 Participants

A total of 28 participants (15 identified as male, 12 as female, and one as other) with ages ranging from 18-62 years old ($M = 27.9$, $SD = 8.5$) were recruited. The participants were, on average, moderately familiar with VR technology and had low to moderate experience with using VR technology.

2.2 Apparatus

For this study, an Oculus Quest 2 HMD was utilized for the VR portion. A 24-inch, 1920×1080 resolution Dell desktop monitor, Logitech mouse, and keyboard were used for the study's computer monitor portion.

The XR Vest consisted of three components: 1) a polyester vest with Velcro sides, 2) a halved Polyvinyl chloride (PVC) pipe tablet mount, and 3) a 10-inch x 6.5-inch tablet. The PVC tablet mount was secured to the back of the vest by zip ties, and the tablet fits securely in the groove of the tablet mount. The Oculus Quest 2 HMD was used with the XR Vest and was screencast to the tablet. The setup enabled the participant to observe the tablet without obstruction (Figure 1).

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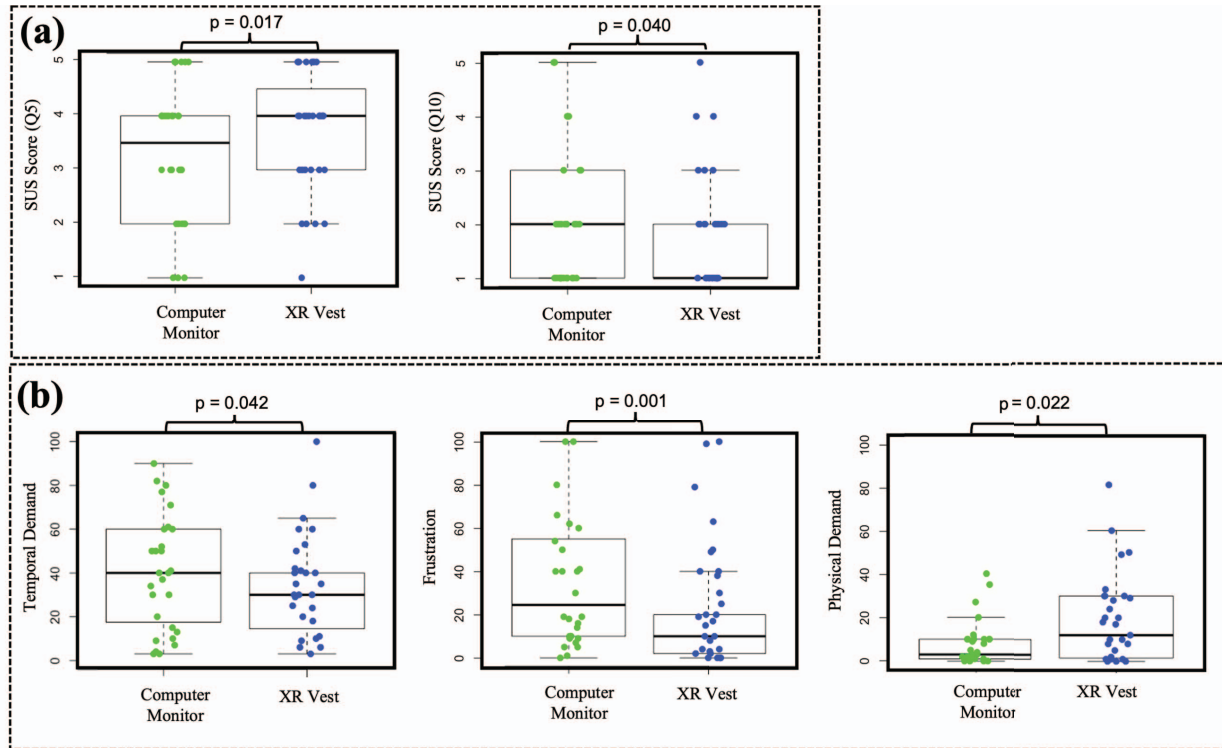


Figure 2: Comparative results between the computer monitor and the XR Vest from the (a) SUS and (b) NASA-TLX

2.3 Experimental Design

Prior to the start of the experiment, the participants were informed about the details of the study and asked to sign a consent form. Then, they completed a demographic survey, which provided information on their prior experience and knowledge of XR technology. Next, the subjects completed a practice session to become familiar with the structure of the training. For the study, three VR training programs were used from Digital Engineering and Magic¹: 1) Feeder Protection Relay Training for Operators (FPR), 2) Occupational Safety and Health Training for Electricians (OSH), and 3) Transformer Oil Sampling in Virtual Reality (TOS). While all the training programs instructed viewers on occupational safety hazards, the topics varied. The FPR training demonstrated the purpose and operation of feeder protection relay devices for substation operators. The OSH training showed the importance of maintaining safety at the workplace while performing maintenance on energized equipment. Finally, the TOS training provided information for electrical utility workers on performing a transformer oil sampling procedure. Each of these programs was experienced through the interfaces (i.e., watching a recording on the desktop computer monitor or observing the program demonstrated through the XR Vest). Each program was randomly assigned to a different interface, and the order of the interfaces was randomized between participants.

2.4 Measures

2.4.1 System Usability Scale (SUS)

The System Usability Scale (SUS) questionnaire measured the usability of the systems (i.e., computer monitor and XR Vest). This was measured using ten statements on a 5-point scale from strongly agree (5) to strongly disagree (1) [1].

¹<https://www.digitalengineeringmagic.com/>

2.4.2 NASA-Task Load Index (NASA-TLX)

Following the VR training program, the participants were asked to fill out a NASA-TLX questionnaire that measured their perceived workload of the task they had just completed. On a scale of 1 to 100, the participant was asked to rate the task on six criteria: Mental Demand, Physical Demand, Temporal Demand, Effort, Performance, and Frustration [3].

2.4.3 Open-Ended Questionnaire

After the participant completed the last interface and the associated questionnaires, they were asked to fill out a short open-ended survey of three questions (Table 1).

Table 1: Open-ended questions

Q1) What areas/fields do you think would benefit from a portable and wearable interface (XR Vest) for demonstration-based training tasks?
Q2) Do you think the XR Vest would be a viable solution for portable and wearable demonstration-based training tasks? Please explain.
Q3) Which type of interface do you mainly prefer (Computer Monitor or XR Vest)?

3 RESULTS

We used a linear mixed-effect analysis to compare the effectiveness of the systems (computer monitor and XR Vest) and training applications (FPR, OSH, TOS) on cognitive load and system usability. Four subjects were removed due to glitches with the XR Vest system that likely impeded learning, cognitive load, and system usability

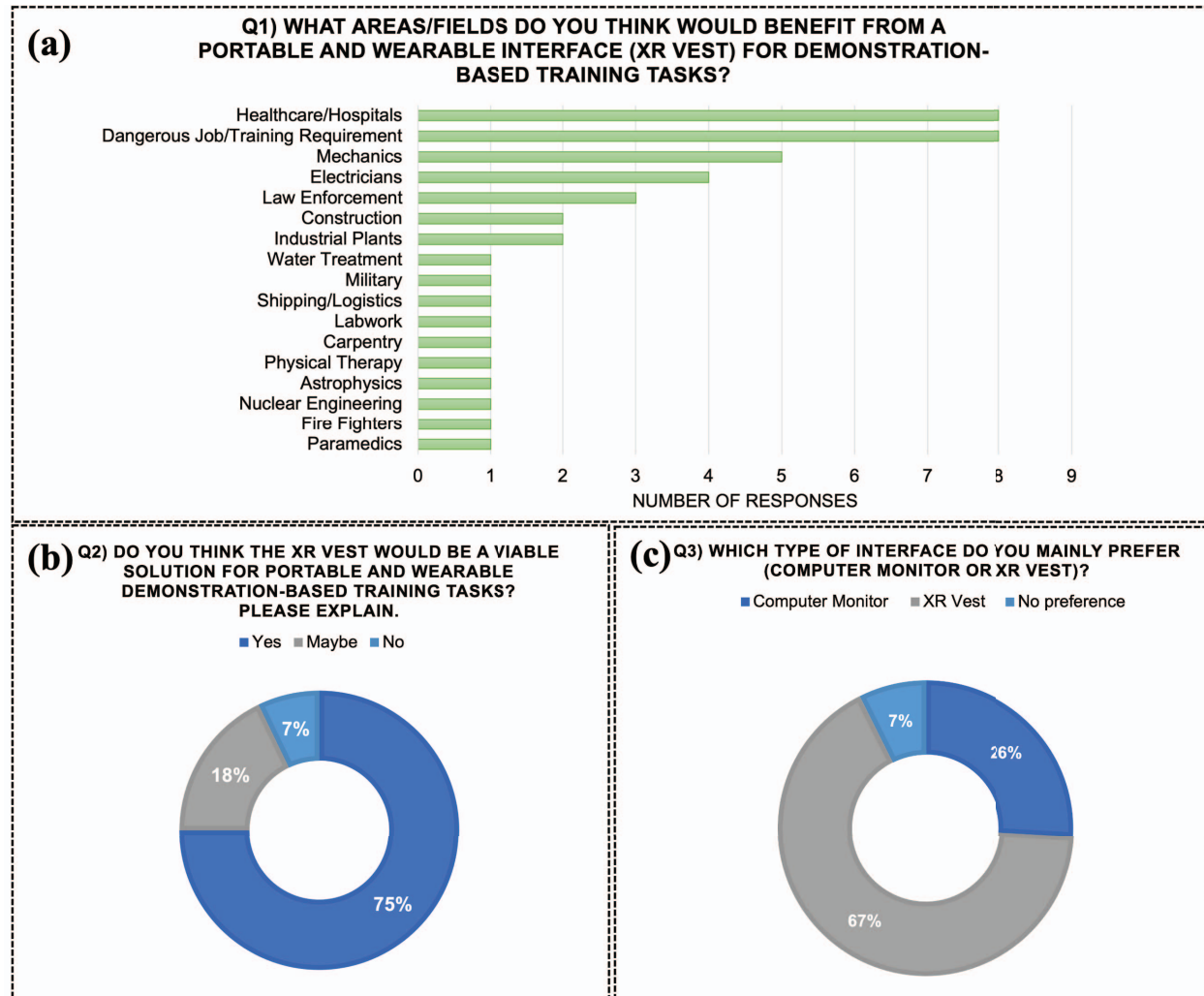


Figure 3: Keyword results from the open-ended survey.

ratings. Outliers were also removed before analysis. The analysis was performed on each question of the SUS and NASA-TLX questionnaires.

3.1 System Usability Scale (SUS)

The results of the SUS (Figure 2a) analysis showed a significant difference between the systems (i.e., the computer monitor and XR Vest) in question 5 (*I found the various functions in this system were well integrated.*) ($p = 0.017$) and question 10 (*I needed to learn a lot of things before I could get going with this system.*) ($p = 0.040$).

3.2 NASA-TLX

The NASA-TLX (Figure 2b) revealed that there were significant differences between the systems in terms of temporal demand ($p = 0.042$), frustration ($p = 0.001$), and physical demand ($p = 0.022$).

3.3 Open-ended Survey

From the open-ended survey answers (Table 1), keywords were extracted from each question. For question 1, the keywords were tallied, whereas for questions 2 and 3, the responses were measured as percentages.

The participants were asked to list areas/fields they felt would benefit from using a portal interface for demonstration-based training. Keywords from the responses were extracted, grouped, and tallied (Figure 3a). Most of the participants' responses were within the fields of healthcare and dangerous jobs. Next, the keywords "yes," "no," or "maybe" were evaluated, which responded to the viability of XR Vest as a portable DBT interface (Figure 3b). All the responses contained one of these keywords. The results show that 75% of participants believed that the XR Vest is a viable solution for DBT. Finally, the keywords "Computer Monitor," "XR Vest," and "no preference" were extracted (Figure 3c). All the responses contained one of these keywords. The results show that 67% of participants preferred the XR Vest as a DBT interfaced compared to the traditional computer monitor platform.

4 CONCLUSION, LIMITATIONS, AND FUTURE STUDIES

This study analyzed the potential usefulness of a novel DBT system. The results from the NASA-TLX survey showed that the XR Vest had an increase in physical demand. This result was expected, as there was movement incorporated in the DBT compared to the stationary setup of the computer monitor. Further, the results showed

that the XR Vest reduced frustration and time pressure on the trainee. The SUS and open-ended surveys showed that most participants preferred the XR Vest for DBT over traditional computer monitor use.

However, a few limitations in this study negatively affected the potential of the XR Vest. Namely, the smaller screen size and the size of the room in which the study was conducted likely impacted the results of this study. Regarding the room, the observing participant could only freely move about one side/wall due to its size and layout. Because of this and the nature of VR programs, the participant could not always observe the XR Vest's screen from an ideal location/angle. Future studies with the XR Vest could be conducted to analyze the effect of different screen sizes in a room that permits greater freedom of movement for the observing participant.

ACKNOWLEDGMENTS

The authors would like to thank Jean Pierre Valentin, Patrick Liam Ferreira, and Sean Mendez, Dr. Michael Brown for their contribution to the design and prototyping of the XR Vest. This material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. 026257-001. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

REFERENCES

- [1] J. Brooke et al. *SUS: A quick and dirty usability scale*, vol. 189 of *Usability Evaluation in Industry*. London, England, 1996.
- [2] R. Grossman, E. Salas, D. Pavlas, and M. A. Rosen. Using instructional features to enhance demonstration-based training in management education. *Academy of Management Learning and Education*, 12(2):219–243, 2013. doi: 10.5465/amle.2011.0527
- [3] S. G. Hart and L. E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. *Advances in Psychology*, p. 139–183, 1988. doi: 10.1016/s0166-4115(08)62386-9
- [4] C. Heyes. Causes and consequences of imitation. *Trends in Cognitive Sciences*, 5(6):253–261, 2001. doi: 10.1016/s1364-6613(00)01661-2
- [5] B. Marques, A. Portela, T. Carvalho, R. Ferreira, D. Siso, M. Marques, D. Almeida, P. Dias, and B. S. Santos. Supporting multi-user co-located training for industrial procedures through immersive virtual reality (vr) and a large-scale display. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 749–750, 2022. doi: 10.1109/ISMAR-Adjunct57072.2022.00158
- [6] M. W. Norris, K. Spicer, and T. Byrd. Virtual Reality: The New Pathway for Effective Safety Training. *Professional Safety*, 64(06):36–39, 06 2019.
- [7] M. A. Rosen, E. Salas, D. Pavlas, R. Jensen, D. Fu, and D. Lampton. Demonstration-based training: A review of instructional features. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 52(5):596–609, 2010. doi: 10.1177/0018720810381071